

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458045.6517 +/- 0.002 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.173 +/- 0.013
Transit depth (Rp/Rs)^2	2.99 +/- 0.45 [%]
Orbital Inclination (inc)	85.06 +/- 0.79
Airmass coefficient 1 (a1)	1.221 +/- 0.014
Airmass coefficient 2 (a2)	-0.171 +/- 0.014
Scatter in the residuals of the lightcurve fit is	1.17 %
Best Comparison Star	None
Optimal Aperture	5.38
Optimal Annulus	23.87
Transit Duration (day)	0.074 +/- 0.0052

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.173 ± 0.013 vs 0.1646 (deviation 0.65σ)
Duration fit vs published	0.074 d vs 0.0758 d (deviation 0.35σ)
Mid-time O–C	2.3 min
Depth SNR	13.3
Residual scatter	1.17 %

Light curve

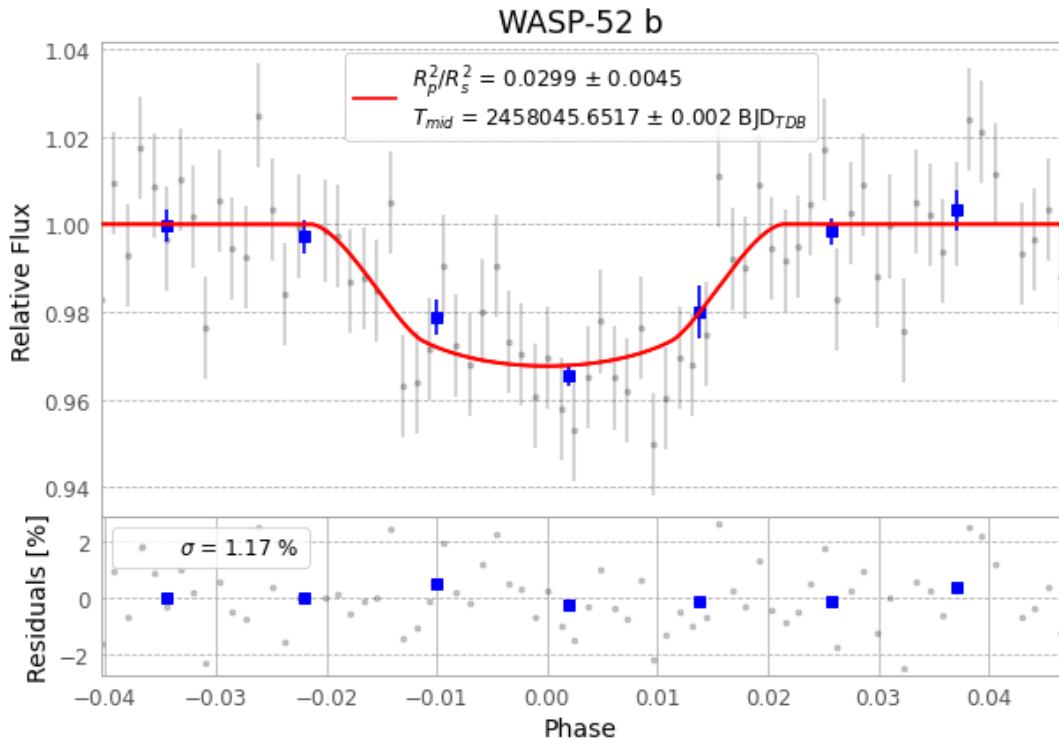


Figure 1 — FinalLightCurve_WASP-52 b_2017-10-18.png (SHA-256 035f98245858761f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [324, 190], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d015c72e76c0a185...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

74 FITS frames (49 MB), WASP-52171019014812.FITS → WASP-52171019052712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-171019.log (3eec522876ec...), FinalLightCurve_WASP-52 b_2017-10-18.png (035f98245858...), FinalLightCurve_WASP-52 b_2017-10-18.csv (c6b2525d3064...)

Compute resources

Wall time 165.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 16:35Z.